

WILLIAM R. JOHNSON

SOFTWARE ENGINEER

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CooCooCaCha
wrjcareer

EXPERIENCE

CancerIQ
2022 - present

Principal Software Engineer
Promoted from Senior Software Engineer in 2026.

Built apps for cancer prevention and early detection using Rails, React, Rust, Elixir, Node.js, and Google Cloud.

- Led the genogram team in rewriting our visual family tree builder in React and refactoring the Rails backend to support new features.
- Researched and developed a deterministic, recursive family tree layout algorithm that preserves paternal/maternal orientation and keeps twins adjacent, replacing our previous Sugiyama-based layout.
- Built a custom query language using parser combinators, enabling product managers to target content based on patient data.
- Worked with a co-founder to standardize an Excel format for patient data and built a Ruby import pipeline using pre-generated UUIDs and transactions to prevent partial patient imports and resume safely after failures. Tested with 1M+ records and 100+ columns.
- Led improvements to our AI-assisted development workflow by grouping apps and services in one folder and writing an AGENTS.md to give AI tools codebase context and development guidelines.

World 50
2018 - 2022

Principal Software Engineer
Promoted from Senior Software Engineer in 2021.

Full-stack engineer and manager of five engineers working on the company's React Native and Node.js platform.

- Rebuilt the legacy application with React Native, sharing most of the code across web, iOS, and Android.
- Rebuilt the API with GraphQL, using DataLoader to avoid N+1 queries.
- Proposed an engineering process, later adopted, in which temporary, cross-functional teams planned and delivered individual epics.
- Set up databases, servers, and build pipelines using AWS, Google Cloud, and CircleCI.
- Built an activity feed that let admins filter activity across the app or open a profile to see one user's history. Wrote a query builder that dynamically unioned activity tables and applied filters within each table and across the combined results.

TSYS
2017 - 2018

Senior Software Engineer
Brought in for React expertise to build banking apps and modernize the team's frontend architecture.

- Rebuilt our shared React component library, removing application-specific business logic and making components more modular to simplify frontend code and improve maintainability.
- Built a dynamic React form system that generated multi-step forms from configuration and constraints supplied by the backend API.
- Introduced modern React patterns, including function components, fragments, and hooks, through a series of team presentations and mentored other developers.

Koble
2016 - 2017

Co-Founder / Lead Engineer
Co-founded a platform for people to offer their skills and discover services in their community.

- Built the React Native app and backend as the sole engineer, owning software architecture and implementation.
- Set up Google Cloud infrastructure, including a Kubernetes cluster, configured CI pipelines, and distributed iOS builds through TestFlight.
- Built a service to automate Let's Encrypt certificate provisioning and renewal for TLS/SSL encryption.
- Built an audit-log service that recorded Firebase change summaries in BigQuery.

Salesloft
2015 - 2016

Full-Stack Software Engineer
Developed features for Salesloft's sales automation platform, from database design and migrations to frontend UI.

- Built a Docker-based QA environment for testing multiple versions of the application simultaneously on the same server cluster.
- Contributed to integrating a third-party sentiment analysis service into the application's email features.
- Learned Ruby, Rails, and Angular on the job with no prior experience in those technologies, contributing my first code on day one.

EDUCATION

Georgia Institute of Technology
Bachelor's Degree - Computer Science
2007 - 2012

SKILLS

Languages: TypeScript, JavaScript, Ruby, Rust, Python

Backend & Data: Rails, Node.js, GraphQL, SQL, PostgreSQL

Frontend & Mobile: React, React Native, HTML, CSS

Cloud & Tooling: AWS, Google Cloud, Kubernetes, Git